



Investigating transformations with GeoGebra

Task A: Using GeoGebra for rotations and enlargements

1) [Task A Question 1](#)

Enlarge object A onto image A'.

- a) What is the scale factor? 4
- b) At what coordinate is the centre of enlargement? (2, 6)

2) [Task A Question 2](#)

Enlarge object B onto image B'.

- a) What is the scale factor? $\frac{1}{2}$ or 0.5
- b) At what coordinate is the centre of enlargement? (2, 4)

3) [Task A Question 3](#)

Rotate object C onto image C'.

- a) What is the angle of rotation? 90° clockwise or 270° anticlockwise
- b) At what coordinate is the centre of rotation? (6, 6)

4) [Task A Question 4](#)

Use the slider to rotate object D onto image D'.

- a) What is the angle of rotation? 130°



Task B: Using GeoGebra for reflections and translations

1) [Task B Question 1](#)

Draw a line through points P and Q.

Reflect object A over the line.

Move the line of reflection down onto each of the numbered lines.

Which line reflects object A onto image A'?

Line 4

2) [Task B Question 2](#)

Reflect object B over the line.

Move point P to each of the number points to turn the line.

At which point does the line reflect object B onto image B'? *Point 6*

3) [Task B Question 3](#)

Translate each of the objects A to E using vectors 1 to 9.

Which vector translates each object onto the image?

a) Vector 6 translates object A onto the image.

b) Vector 4 translates object B onto the image.

c) Vector 9 translates object C onto the image.

d) Vector 8 translates object D onto the image.

e) Vector 3 translates object E onto the image.

Task C: Using GeoGebra to identify transformations

[Task C](#)

Find and describe each transformation below.

1. $A \mapsto B$ *Rotated 90° clockwise around $(0, 1)$.*

2. $A \mapsto C$ *Translated $\begin{pmatrix} 9 \\ -1 \end{pmatrix}$*

3. $A \mapsto D$ *Reflected over the x-axis.*

4. $A \mapsto E$ *Enlarged by a scale factor of 3 from $(-6, 6)$.*

